

1) What is Statistics?

Ans : Statistics is the science of collecting, organizing and analyzing data.

2) Define the different types of statistics and give an example of when each type might be used.

Ans : **1. Descriptive Statistics**

Descriptive statistics is used to summarize, organize, and describe data.

2. Inferential Statistics

Inferential statistics is used to **make predictions or draw conclusions**

3) What are the different types of data and how do they differ from each other? Provide an example of each type of data.

Ans : **Qualitative Data - qualities or categories**

Nominal Data - Data with **no order or ranking**

E.x. Gender, color

Ordinal Data - Data with **order or ranking**

E.x. Class ranks, Education level

Quantitative Data - measured in numbers

Discrete Data - **countable values**

E.x. Number of cars, Number of books

Continuous Data - **any value including decimals within a range.**

E.x. Height, Weight

4) Categorise the following datasets with respect to quantitative and qualitative data types:

(i) Grading in exam: A+, A, B+, B, C+, C, D, E

(ii) Colour of mangoes: yellow, green, orange, red

(iii) Height data of a class: \[178.9, 179, 179.5, 176, 177.2, 178.3, 175.8,...]

(iv) Number of mangoes exported by a farm: \[500, 600, 478, 672, ...]

Ans : i) ordinal

ii) nominal

iii) counties

iv) discreet

5) Explain the concept of levels of measurement and give an example of a variable for each level.

Ans : nominal scale data, ordinal scale data, Interval scale data, ratio scale data

nominal scale data e.x. gender , color

ordinal scale data e.x. Class rank

Interval scale data e.x. Temperature

Ratio scale data e.x. Hight, Weight

6) Why is it important to understand the level of measurement when analyzing data?

Provide an example to illustrate your answer.

Ans: is important because it determines the appropriate statistical methods, calculations, and interpretation of data, ensuring accurate and meaningful results.

7) How nominal data type is different from ordinal data type.

Ans : nominal data type has no order and ranking or ordinal data type has order and ranking

8) Which type of plot can be used to display data in terms of range?

Ans : box plot

9) Describe the difference between descriptive and inferential statistics. Give an example of each type of statistics and explain how they are used.

Ans : **descriptive statistics** is used to summarize, organize, and describe data

They are present data in simple form

Inferential statistics is used to **make predictions or draw conclusions**

They are used To make decisions

10) What are some common measures of central tendency and variability used in statistics?

Explain how each measure can be used to describe a dataset.

Ans : mean (average) give over all average of the data

Median ordered dataset middle value

Mode one dataset most frequently data